Math 252Z [redacted] **Written Homework - 1.5**

1. Evaluate the following integrals by applying the formulas in Lecture 1.5 Example 6.

1). $\displaystyle\int \cos(4x)\,dx =$

2). $\displaystyle\int e^{-2x}\,dx =$

3). $\displaystyle\int \sin\left(\frac{x}{2}\right)\,dx =$

4). $\displaystyle\int \frac{1}{4+3x}\,dx =$.

2. (1.5) Evaluate the following integrals by substitutions. *Make sure to indicate the u-function.*

1). $\displaystyle\int x^2\sqrt{x^3+1}\,dx$

2). $\displaystyle\int \frac{\sec^2(1/x)}{x^2}\,dx$

3). $\displaystyle\int e^x\sin(e^x)\,dx$

4). $\displaystyle\int \frac{x}{x^2+1}\,dx$

5). $\displaystyle\int (1+\tan\theta)^5 \sec^2\theta \, d\theta$

6). $\displaystyle\int \frac{e^x}{e^x+1}\, dx$

7). $\displaystyle\int \frac{\sin x}{1+\cos^2 x}\, dx$

8). $\displaystyle\int x^5(x^6+10)^{10}\, dx$

9). $\displaystyle\int \frac{\cos\theta}{\sin^3\theta}\,d\theta$

10). $\displaystyle\int \frac{1}{x\ln x}\,dx$

11). $\displaystyle\int \sin(\tan\theta)\sec^2\theta\,d\theta$

12). $\displaystyle\int_0^1 \frac{(\arctan x)^2}{1+x^2}\,dx$

13). $\displaystyle\int_0^{\pi/3} \sec^3\theta \tan\theta \, d\theta$

14). $\displaystyle\int_1^2 x\sqrt{x-1}\, dx.$

15). $\displaystyle\int x^3(2+x^4)^5 \, dx$

16). $\displaystyle\int \cos^3\theta \sin\theta \, d\theta$

17). $\displaystyle\int \frac{(\ln x)^2}{x}\,dx$

18). $\displaystyle\int_0^{1/2} \frac{\sin^{-1} x}{\sqrt{1-x^2}}\,dx$